# Sound and Music Conceptual understandings

Discuss the electromagnetic spectrum

Compare longitudinal and transverse waves.

Draw and label a sound wave including displacement, amplitude, frequency and wave length

Draw various sound waves and label them as high pitch and low amplitude, Low pitch and high amplitude, and high pitch high amplitude and low pitch low amplitude

Compare how sound travels through solids, liquids and gases.

Compare how sound reflects of different surfaces.

Describe an echo and several uses for echoes

Compare the speed of light with the speed of sound.

Describe how you can calculate how far away a lightning strike is by the timing of the thunder. Provide a formula for this.

Describe the importance of sound in communication

Recognise the use of sign language and common hand signs when hearing is inefficient

Draw and label a diagram of the ear and provide a function for each labelled part of the ear.

Compare the role of the cochlea with the role of the semicircular canals.

How do vocal cords produce speech?

Discuss the threshold of hearing for loudness and frequency

Discuss the common causes of hearing loss

Be familiar with devices that help people hear such as hearing aids and cochlea implants

Explain the difference between music and noise.

Compare how a trumpet and drum produce music.

Explain ways to transmit sound over larger distances than is normally possible with shouting.

Investigate the principles of recording and playing back sound.

Construct a musical instrument

Explain the concept of building acoustics.

Describe the behaviour of waves, reflection, transmission, absorption, and diffusion.

Describe what a decibel is and how the logarithmic decibel table works.